

SANYAM KAUL

kaulsanyam@gmail.com — +91-7018498338

LinkedIn: <https://www.linkedin.com/in/kaulmesanyam/>

GitHub: <https://github.com/kaulmesanyam/> — LeetCode: <https://leetcode.com/u/kaulmesanyam/>

PROFILE SUMMARY

Software Engineer with an M.Tech in Computer Science from IIT Hyderabad and 5 years of experience in building scalable production systems and applied AI. Skilled in architecting RAG-based LLM systems, optimizing distributed data pipelines, and developing high-performance financial platforms. Proven track record of reducing cloud costs by 15% and optimizing system latency by 6x. Expert in bridging the gap between core backend engineering and applied AI research.

TECHNICAL SKILLS

- **Languages:** Java, Python, SQL, C++, Golang, Rust, Javascript, Shell/Bash.
- **AI/ML:** LLMs (RAG, Fine-tuning, PEFT, LoRA/QLoRA), Transformers, LangChain, LangGraph, VAE/Anomaly Detection, Reinforcement Learning (RLHF), MCP.
- **Backend:** Spring Boot, FastAPI, REST, gRPC, GraphQL, Node.js, Microservices.
- **Data & Infrastructure:** Cloud (AWS, Azure), EKS, Lambda, SQS, SNS, S3, Docker, Kubernetes, Kafka, Redis.
- **Databases:** PostgreSQL, MongoDB, ChromaDB, Elasticsearch.
- **Tools:** Git, Jenkins, GitHub Actions, CI/CD, Grafana, Datadog, Prometheus.

EXPERIENCE

Netradyne

Data Engineer (ML Systems)

Bengaluru, India

Jul 2025 - Present

- Architected AI-powered agent for automating Root Cause Analysis workflow using LangChain, LangGraph, and custom MCP servers, reducing analysis time by **60%**.
- Implemented prompt chaining and conditional logic achieving **40% reduction in hallucinations** and improved context retention.
- Integrated agent into VS Code Copilot with AWS S3 file operations and automated report generation.
- Designed and implemented AWS Monitoring System (Python, MongoDB, FastAPI) for machine metrics and costs, resulting in a **10–15% reduction in cloud spend**.
- Optimized Video Reprocessing System using EKS, SQS, SNS, and ASGs, reducing latency from **2 hours to 20 minutes (6x speedup)** for 50K+ video dataset creation.
- Built scalable infrastructure for large-scale video feature engineering and automated ML dataset generation.
- **Tech Stack:** Python, FastMCP, LangChain, LangGraph, FastAPI, MongoDB, PostgreSQL, Redis, AWS (EKS, S3, SQS, SNS, CloudWatch), Dash.

WinesecretHK

AI Architect

Remote

Apr 2024 - Jun 2025

- Built a **RAG-based AI Pairing System** using Python, AWS RDS, and ChromaDB for high-precision flavour-profile matching beyond standard semantic search.
- Developed and submitted a patent for a **HyDE-inspired RAG algorithm** optimized for gastronomical compatibility and complex pairing logic.
- Implemented multi-modal agentic workflows using LangChain and LangGraph to handle complex, multi-step user interactions and intent recognition.
- Integrated Stripe payment systems and POS webhooks to support seamless ordering and transaction workflows.
- **Tech Stack:** Python, FastAPI, AWS RDS, ChromaDB, LangChain, LangGraph, Stripe.

BlackRock

Software Development Engineer 1

Gurugram, India

Jan 2021 - Jun 2023

- Developed full-stack Financial Information Reporting Tool using Spring Boot and Angular, reducing client report setup time from **1.5 months to 2-3 days**.
- Standardized REST and GraphQL API workflows across **50+ global teams**, significantly improving system reliability and developer productivity.
- Built an **Azure DevOps** analytics dashboard to track CI/CD performance and improved backend test coverage to **85%+**.
- **Tech Stack:** Java (Spring Boot), Angular, REST, GraphQL, Golang, **Azure DevOps**.

PROJECTS

Car Insurance OCR (Structured Data Extraction) — AI Developer

POC

- Developed a Python-based pipeline to extract structured data from unstructured insurance PDFs using Tesseract OCR and OpenAI LLMs.
- **Tech Stack:** Python, OpenAI APIs, Tesseract OCR, FastAPI.

Fraud Detection using Representation Learning — Machine Learning Engineer

[GitHub](#)

- Built a Variational Autoencoder (VAE) model for credit card fraud detection on imbalanced datasets, achieving **91%+ accuracy**.
 - **Tech Stack:** Python, PyTorch, VAE, Scikit-Learn.
- LooksMaxing AI Platform** — *Full Stack AI Developer* [Live Link](#)
- Developed an AI-driven personalization platform using Gemini and prompt engineering for virtual try-on and recommendation workflows.
 - **Tech Stack:** Python, FastAPI, Gemini API, Prompt Engineering.

EDUCATION

IIT Hyderabad
M.Tech in Computer Science

Hyderabad, India
2025

- CGPA: 8.90 — Specialized in Distributed Systems and Applied AI.

VIT-AP University
B.Tech in Computer Science (Networks & Security)

Amaravati, India
2021

- CGPA: 8.90 — Focus on Cybersecurity and Network Infrastructure.

RESEARCH & ACHIEVEMENTS

- **M.Tech Thesis:** Optimized HyperLedger Fabric blockchain architecture to reduce node failure rates; presented at **Indosys 2025**. [arXiv](#)
- **1st Place (IITH FoML):** Developed a winning ensemble model (Random Forest/GBDT) for predictive analytics, achieving 98% accuracy in predicting the correct user class on the test set.
- **15th/100+ (IITH DL):** Ranked in the top 15 in Deep Learning competition using ResNet and ViT architectures, achieving 92% accuracy in labeling image data on the test set.
- **Amazon AI Hackathon:** Developed a Vision-Language Model (VLM) based system to extract product data from images, featuring a chat-based UI for users to query product information.
- **Deep Learning Architectures:** Implemented CNN, RNN, and Transformers from scratch using NumPy to deepen core algorithmic understanding.